

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

10th March 2022

Subject CB1 – Business Finance

Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)

Total Marks: 100

- Q. 1)** Which of the following is not an objective of an employee:
- a. Job security
 - b. Training
 - c. Ethical marketing of good
 - d. Safe working conditions
- [2]**
- Q. 2)** In circumstances where a restriction has been placed on the evidence that the auditor can access or where the auditor disagrees with the treatment of a matter related to company accounts, which of the following does the auditor issue:
- a. Disclaimer of an opinion
 - b. Qualified opinion
 - c. Reserved opinion
 - d. Adverse opinion
- [2]**
- Q. 3)** Mr. X started his own event management company and hired his friend Mr. Y to become the Creative Head. Then, they both hired 10 more employees to help them scale the event management business. What kind of business entity is this?
- a. Sole Trader
 - b. Partnership
 - c. Limited Company
 - d. Limited Liability Partnership
- [2]**
- Q. 4)** The difference between gross investment & net investment is
- a. Capital investment
 - b. Net sale of investments
 - c. Depreciation
 - d. Revaluation reserve
- [2]**
- Q. 5)** Which of the following is the correct sequence of finance options in increasing order of risk:
- a. Subordinate debt, Debenture stocks, Convertible preference shares, Equity
 - b. Equity, Subordinate debt, Debenture stocks, Convertible preference shares
 - c. Convertible preference shares, Equity, Subordinate debt, Debenture stocks
 - d. Debenture stocks, Subordinate debt, Convertible preference shares, Equity
- [2]**
- Q. 6)** Which of the following scenarios best extends itself to JIT inventory management approach of a company?
- a. The company has a complex supply chain with different parts being sourced from multiple countries and the company would like to keep air freight to a minimum
 - b. The company engages in retail of fast moving consumer goods

- c. The company manufactures seasonal but standard items with a local supply base centered around its manufacturing unit
- d. The company engages in cold storage warehousing where it stocks seasonal agricultural products for sale in off season

[2]

Q. 7) Which of these is a bearer document?

- a. Non-recourse factoring
- b. Commercial Paper
- c. Overdraft
- d. Trade credit

[2]

Q. 8) Which of the following is not a part of sustainability report produced by the directors?

- a. Business sustainability related to sales forecasts, threats, opportunities etc
- b. Social sustainability related to labor practices, health & safety, work hours etc
- c. Economic sustainability relating to anti-competitive practices, anti-corruption etc
- d. Environmental sustainability relating to waste disposal, effluents, water consumption etc

[2]

Q. 9) Which of this is not an example of risk mitigation:

- a. Adding a cancellation option in a life insurance policy increases risk; the Company decides to not provide this option
- b. The Company is unsure about lending to a newly launched company. So the Company decides to use non-recourse factoring for all its bills
- c. The Company has a lot of payments to be made next year in Euros, so the Company has purchased an interest rate swap
- d. The Company's Actuary has projected that a major catastrophe event may occur 10 years from now. As the event is far away, the Company decides to not consider it in this planning and growth discussions

[2]

Q. 10) An asset costing Rs.10,000 was purchased 5 years ago & deemed to have a useful life of 10 years, was recently sold for Rs.5,000. Its value in the books at the time of sale was Rs.4,000. Calculate the written down value depreciation rate used for the asset:

- a. 14.86%
- b. 20.11%
- c. 10%
- d. 15%

[2]

Q. 11) Mr. X earned INR 100,000 by trading in shares. He is confused on how to use this money – To fund his son's MBA, who will repay after 2 years, INR 50,000 every year for 3 years or to fund his business expansion where there is 30% chance of losing all the money and 70% chance of getting back INR 200,000 next year. What is the opportunity cost of funding his son's MBA? (Assume discount rate of 10% p.a.)

[5]

Q. 12) Explain how the following items will be handled in books of accounts of a company:

- i) Take over of a loss making entity with negative net worth taken over in an all cash transaction. (1)
 - ii) Legal suit filed over disputed ownership of an asset owned & recorded in the books of accounts. (1)
 - iii) Revaluation of building value. (1)
 - iv) Interest due but not paid on director's loan. (1)
 - v) Software development expenses. (1)
- [5]**

Q. 13) A large scale MNC is head quartered in India with its operations spread out of Dubai, Sri Lanka, Singapore and Thailand. Of its operational countries, it enjoys double taxation relief in Sri Lanka and Singapore. In the last financial year, it earned profits before tax of INR 15 lac in its head quarter and equivalent to INR 5 lac, INR 1 lac, INR 10 lac and INR 3 lac respectively in its branches. Considering tax rates as below, calculate the total tax that the MNC would require to pay in India, after payment of taxes of its branches.

Country	Tax Rate
India	20%
Dubai	5%
Sri Lanka	25%
Singapore	15%
Thailand	30%

[5]

Q. 14) Generate a statement of profit & loss for an insurance company based on the following information:

Particulars	Amount in Rs.
Insurance Premium Earned During the Year	10000
Taxation on Profit	30%
Capital Gains on Assets Held for Capital Adequacy	1500
Capital Loss on Assets Held as Reserves	1500
Investment Income from Assets Held as Reserves	3000
Investment Income from Assets Held for Capital Adequacy	2500
Claim Expenses Incurred	9000
Operational Expenses Incurred	2000

[5]

- Q. 15)** Your brother runs a private limited company. One of his friend suggested to him, now that his Company is growing, he should consider getting a quotation to become a public limited company. Your brother is unsure of what this means for his business and has asked for your advice. Explain to your brother what are the advantages and disadvantages of getting a quotation to make his company public limited.

[5]

- Q. 16)** Calculate total interest payable & interest coverage ratio:

Return on Assets	12.00%	
Gross Margin	30.00%	
Sales	100,000	
Fixed Expenses	15,000	
Equity	50,000	
	Interest Priority Lower Percentile	Interest on Debt
Mortgaged Debt	0.00%	8.00%
Mezzanine Debt	10.00%	10.00%
3rd Priority Debt	40.00%	12.00%

[5]

- Q. 17)** i) Define what is “Margin” in a future’s trade.

(1)

- ii) Explain the 2 types of margin.

(2)

- iii) Suppose you entered into a futures contract to sell a bond 15 days later at INR 1,000. The clearing house has asked both of the parties to pay INR 100 at the start of the contract. Considering that the price of the bond in the next 15 days is as below, what is the total net amount you pay to the clearing house? Show workings for each day over this period.

(2)

Day	Price of the bond
0	900
1	1050
2	1000
3	950
4	850
5	700
6	800
7	900
8	1050
9	1000
10	1100
11	1150
12	1200
13	1100
14	1050
15	950

[5]

Q. 18) Generate a cash flow statement from the following entries:

Particulars	2020-21 Amount Rs.	2021-22 Amount Rs.
Liabilities		
Share Capital	270,000	390,000
Reserves & Surplus	144,000	252,000
Trade Payables	52,200	66,000
Assets		
Fixed Assets	280,200	498,000
Inventories	66,000	78,000
Debtors	108,000	117,000
Cash	12,000	15,000
Total	466,200	708,000
Additional Info:		
Depreciation	60,000	
Income Tax Paid	15,000	
Share Premium Generated	48,000	

[5]

Q. 19) ABC Ltd. is an insurance company aiming to significantly grow in the next year.

- i) What may be the reasons for aiming to grow? (1)
- ii) List the constraints in achieving this growth. (2)
- iii) What are the methods for ABC Ltd. to achieve growth and what are their advantages? (4)
- iv) ABC Ltd. has chosen to acquire its competitor XYZ Ltd. in order to grow. For the purpose of acquisition, ABC Ltd. is trying to evaluate the appropriate purchase price for the target company. Considering that XYZ Ltd. is expected to have a future stream of profits as below, calculate the appropriate purchase price @ 10% risk free rate of return. You may ignore any terminal value of the target company.

Year	Profit in Cr.
1	-6
2	-2
3	1
4	3
5	10

(3)

- v) List the other considerations, besides price that ABC Ltd. should evaluate. (3)
- vi) ABC Ltd. does not have enough funds required above to acquire XYZ Ltd. Name any 4 options to arrange for funds. (2)

vii) If ABC Ltd. decides to raise equity while its beta is 1.5 and risk premium is 5%. What would be the revised cost of capital of ABC Ltd.? (1)

viii) For finalizing on this acquisition deal, ABC Ltd. consulted its financial manager. What are the considerations to be made by him to decide on this acquisition? (4)
[20]

Q. 20) i) An automotive company has only a half of the receivable turnover ratio & inventory turnover ratio while having double the creditor turnover ratio of its competitors. Discuss how you will analyze each of the components, their measurement approaches & the overall impact on financing, profitability & brand image. (15)

ii) The company wants to prepare a budget for the next financial year. What limitations of budgeting should the company be aware of before proceeding? (5)
[20]
